Kunvar Kanhaiya

PhD - Computer Science and Engineering
Computer Networks and Network Monitoring
Indian Institute of Technology Indore

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Academic Website

Research Summary

My PhD research at IIT Indore focuses on network monitoring, sketch-based data structures, packet scheduling, traffic analysis, machine learning for the network data plane, and privacy-preserving communication. My work aims to develop scalable and memory-efficient algorithms for high-speed networks, improving measurement accuracy, scheduling efficiency, and network performance while reducing computational and memory overhead. I explore practical algorithmic solutions for efficient traffic measurement, intelligent packet scheduling, and secure communication.

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
PhD	Indian Institute of Technology Indore	7.07 (Current)	2022-Present
M.Tech	National Institute of Technology Goa	6.41	2020
B.Tech	Rajkiya Engineering College Akbarpur Ambedkar Nagar U.P.	66.94%	2018
Senior Secondary	St. Francis Inter College Nigohan Lucknow	78.20%	2013
Secondary	St. Francis School Nigohan Lucknow	65.33%	2011

Teaching Experience

- **Parul University** *2021–2022*
  Assistant Professor Gujarat, India
- **ITM GIDA** *2020–2021*
  Assistant Professor Gorakhpur, Uttar Pradesh, India

Research Supervision

- **B.Tech. Project Supervision** – Guided 2 B.Tech. students on projects related to privacy-preserving communication.
- **M.Tech. Research Mentoring** – Mentored 1 M.Tech. student in research on sketch placement for efficient network monitoring.

Fellowships & Awards

- **C-DOT Fellowship** – Centre for Development of Telematics (C-DOT) *2024–2028*
  Doctoral fellowship supporting research in computer networking and telecommunications.

Publications

- **PackMorph: Intelligent Traffic Signature Morphing for Privacy-Preserving Communications**
  Kunvar Kanhaiya, Sourabh Singh, Pralhad Magadum, Aniruddha Singh Kushwaha
  IFIP Networking Conference, 2026 IEEE Xplore
- **Quiver: Dynamic Quantile-Assisted Scheduling to Reduce Rank Inversion**
  Kunvar Kanhaiya, Sourabh Singh, Pralhad Magadum, Rituraj Patel, Aniruddha Singh Kushwaha
  IFIP Networking Conference, 2026 IEEE Xplore
- **SketchPlacement: Selective Count-Min Sketch Deployment for Efficient Network Flow Monitoring**
  Sandip Ambre, **Kunvar Kanhaiya**, Sourabh Singh, Aniruddha Singh Kushwaha
  IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS), 2026 Accepted
- **Enabling Seamless Integration of ML-Based Network Functions into the Network Dataplane**
  Sourabh Singh, **Kunvar Kanhaiya**, Pralhad Magadum, Rituraj Patel, Aniruddha Singh Kushwaha
  IEEE Networking Letters, Vol. 8, pp. 82–86, 2025 IEEE Xplore
- **A Primitive Based Machine Learning Framework for the Network Data-plane**
  Sourabh Singh, **Kunvar Kanhaiya**, Pralhad Magadum, Aniruddha Singh Kushwaha, Ayan Mondal
  IEEE ANTS, pp. 1–6, 2024 IEEE Xplore
- **Machine Learning with Python (Book Chapter)**
  Pralhad Magadum, **Kunvar Kanhaiya**, Aniruddha Singh Kushwaha
  In: Recent Advances in Simulation and Modeling, Apple Academic Press / CRC Press, 2026 Publisher

Professional Service

- **Reviewer**, IEEE International Conference on Communications (ICC) *2025*
- **Reviewer**, 4th International Conference on Data Science and Network Engineering (ICDSNE), NIT Agartala *2026*

Research Projects

- **Efficient Network Monitoring Using Sketch-Based Algorithms** *2022–Present*
  PhD Research | Dr. Aniruddha Singh Kushwaha
  – Developed memory-efficient sketch-based approaches for network traffic measurement and flow monitoring.
  – Investigated selective Count-Min Sketch deployment to improve measurement efficiency and reduce memory overhead.
  – Explored efficient packet scheduling, traffic analysis, and ML-based network data-plane processing.

- **Range-Free Localization in Wireless Sensor Networks** *2019–2020*
  Dr. Keshavamurthy B.N.
  – Implemented range-free localization techniques to estimate the positions of unknown sensor nodes using beacon nodes with known locations.
  – Developed and evaluated the localization approach using MATLAB.

- **Fire Fighting Robot** *2020*
  ITM College Project
  – Developed an Arduino Uno-based firefighting robot utilizing IR sensors for fire detection.

Technical Skills

- **Programming Languages**: Python, C/C++
- **Networking and Data Plane**: P4, Wireshark, NS-3
- **Hardware Description Languages**: Verilog, VHDL
- **Tools and Databases**: MATLAB, MySQL

Key courses taken

- **Mathematics**: Linear Algebra, Probability & Random Processes, Algorithmic Graph Theory, Discrete Maths,
- **Networks**: Computer Network, Advanced Computer Network, Wireless Networks and Applications
- **Data Science**: Machine Learning, Pattern Recognition

Positions of Responsibility

- **Overall Coordinator**, Research Symposium 4.0, IIT Indore. *2026*
- **Overall Coordinator**, Research Symposium 3.0, IIT Indore. *2025*
- **Co-Coordinator**, Research Symposium 2.0, IIT Indore. *2024*
- **Coordinator**, Research Symposium, IIT Indore. *2022 - 23*
- **Volunteer**, MIKE International Conference, NIT Goa *2019*
- **Co-Coordinator**, KSHITIZ'17, REC Ambedkar Nagar *2017*

Achievements

- **JEE Main**, Cleared JEE Main *2013*
- **JEE Main**, Cleared JEE Main *2014*
- **GATE**, Cleared GATE *2018*